University of East Anglia Norwich

School of Computing Sciences

ACADEMIC YEAR	2026-2027
MODULE	CMP-6057A ADVANCED WEB DEVELOPMENT
ASSESSMENT NAME	Coursework 2
WEIGHTING	40%
SET BY	Dr Cheng Wang
CHECKED BY	Dr Jeannette Chin
RELEASE DATE	3 November 2026
SUBMISSION DATE	18 January 2027 at 15:00
SUBMISSION LOCATION	Blackboard and CMP Stugit
FEEDBACK RETURN DATE	15 February 2027

OVERVIEW & LEARNING OUTCOMES

Learning Outcomes Assessed:

Web Programming

- able to use modern browser technologies and pretrained AI models to create an interactive WebAI application
- able to design and evaluate a web application that integrates browser inference with a local AI service through a clear, safe interface

Professionalism

- able to make justified technical choices, test limitations, communicate results and demonstrate independent learning
- able to practice and enhance employability skills through web development, presentations and demonstrations.

TASK SPECIFICATION

Task Spec / Core Objectives:
Overview

For this individual assignment, design and build a Local Feedback Intelligence Assistant. Your application will analyse public app-review data with a pretrained Transformers.js model and use a lightweight local language model, accessed through Ollama, to propose a bounded follow-up action. A person must review the proposal before it can be accepted.

The coursework assesses WebAI implementation. You may use techniques from the first half of the module, but no particular framework, authentication system, or database is required or separately marked. You will build the application from the ground up. No starter application is provided.

Model choice. *The recommended baseline is Xenova/all-MiniLM-L6-v2 in Transformers.js and gemma3:1b in Ollama. These are not compulsory. Suitable alternatives are welcomed when they meet the capability, privacy and reproducibility requirements below. Choosing a larger or less common model does not itself earn more marks.*

Description

Build a web application that helps users explore and triage app reviews.

Public dataset:

Use the English classification data from Zero-shot Bilingual App Reviews Mining with Large Language Models (Zenodo record 11066414). Select an English file from dataset/for_classification/.

Core Tasks:

- Import and validate the user feedback data. Show its source, usable row count and clear errors.
- Present feedback in a usable interface that supports selection and inspection.
- Use a Transformers.js-compatible embedding model in the browser to implement one meaningful semantic feature. Examples include intent classification, semantic search/ranking, or related/duplicate-feedback discovery.
- For selected feedback, send the text and relevant semantic evidence to one lightweight model running locally through Ollama.
- Display one closed, structured proposal containing at least an allowlisted action, a short rationale, source-evidence reference(s), and a confidence or uncertainty indicator. Validate it locally and require an explicit confirm or cancel decision before updating the interface.
- Demonstrate and record test results for the semantic feature and the complete proposal flow, including successful, ambiguous and failed cases. Clearly state the desirable features you implemented in a single page user guide.

Model and runtime requirements:

- Browser model: must run through Transformers.js from ONNX weights and support the chosen semantic feature without a paid API key.
- Local model: must run through Ollama locally and reliably attempt the chosen structured contract.
- Documentation: state the exact model ID/tag and version, licence, approximate download size, test hardware and reasons for each choice.
- Reproducibility: record preprocessing, pooling/normalisation where relevant, prompts, schema, thresholds and any fixed sample or seed.

Bounded-agent requirements:

- Treat imported feedback and model output as untrusted data.
- Restrict proposals to a small documented set of actions and validated arguments. The local model may propose only actions defined by the application. It must not call a tool outside the documented allowlist.
- Show the proposed action and evidence before a human confirms or cancels it.
- Reject malformed or disallowed output safely and provide a recoverable response when a model or service is unavailable.

P1. Essential Features (50 marks):

Your application must support a complete flow from importing data to reviewing a proposed action. Marks reward correctness, relevance, usability and technical reasoning rather than the use of a particular web framework.

- **Application and data journey (10 marks)**
  - Import, validate and display attributed public data through a clear review flow.
- **Transformers.js semantic feature (20 marks)**
  - Run embedding inference in the browser, use the embeddings for the chosen semantic feature, present interpretable evidence and handle ambiguous results appropriately.
- **Bounded local-agent (15 marks)**
  - Produce a useful structured proposal, validate its action and arguments, require user confirmation and handle invalid output or unavailable services safely.
- **Reliability, usability and code quality (5 marks)**
  - Provide clear status and error messages, a usable interface, maintainable code and reproducible setup instructions.

Minimum requirement. *P1 cannot be fully met unless the submission demonstrates both a working Transformers.js semantic feature and a locally generated proposal that requires user confirmation.*

P2. Desirable features (25 marks):

Develop the application beyond the essential requirements by adding one substantial extension or a small group of related features. Possible directions include:

- Richer semantic analysis: add abstention, semantic search or clustering, retrieval-grounded evidence, or a comparison of embedding approaches.
- Stronger agent behaviour: support several allowlisted proposal types, an editable plan, staged confirmation or reversible actions.
- Tool or MCP integration: add one relevant read-only or reversible capability with argument validation and explicit user confirmation.
- Safety and reliability: test prompt injection, maintain an audit history, handle timeouts and retries, repair invalid structured output, or test adversarial cases.
- Interaction and explanation: show source evidence and similarity information, allow proposals to be edited, improve accessibility, or handle low-confidence results clearly.

Note: *P2 is assessed by the quality, relevance and evaluation of the extension, not by the number of features implemented. The examples above are illustrative, not a checklist.*

P3. In-person VIVA Defence (25 marks)

You will be allocated a slot of 10 minutes in the week commencing on Jan 18th, 2027 to defence your work. This session is compulsory. You must bring your project code with you on the day and ready to run your application in your allocated slot. Absence and without EC approval will result in a mark of zero being awarded for this assessment.

- The defence may include but not limited to:
  - Live VIVA: You will be asked 2 random questions about your implementation.
  - Live Code Modification: During your assessment session, you may be asked to make a simple 5-minute code adjustment.

Relationship to formative assessment

Lectures and laboratories introduce browser inference, Transformers.js, semantic representations, local models and structured output, bounded agents and tool integration through independent examples. They support the required knowledge but do not provide a staged coursework solution.

DELIVERABLES

1. **Recorded demo video** (.mp4) with no more than 10 minutes to be submitted to Blackboard.
    - Show the application running from import to confirmed and cancelled proposals, including one safe failure or rejected output.
    - Identify the browser and local models, explain the semantic evidence and validation boundary, and demonstrate any P2 features for which you claim marks.
    - Include concise critical reflection. Do not rely on slides in place of demonstrating the genuine application.
2. **Source code submission**

- Submit the complete source code to the CMP Stugit repository before due date, including configuration files but do not include node_modules, builds, model caches, the full dataset and secrets.
- A README file with concise install/run steps; exact model and architecture; data/model attribution; semantic method; proposal structure; action validation; limitations.
- An AI_USAGE.md file detailing permitted AI-tool use and how outputs were integrated.

Note: *Source code does not carry a separate mark, but markers will inspect it to verify the demonstrated features and assess code quality. Missing, inaccessible or non-runnable source code may prevent marks from being awarded for features that cannot be verified.*

3. **A single A4 page user guide** (filename must be userguide.docx). Provide the URL to your CMP stugit repo in this user guide, and clearly state the desirable features you have implemented.

Note: *this guide will not be marked but it will help the markers to locate your source code in Stugit.*

4. **Compulsory in person VIVA.** Each person will be given a 10-minutes slot to defend their work submitted. Exact time slot will be advised nearer the time. This in-person viva attendance is mandatory. Failure to attend and without EC approval will result in a mark of zero being awarded to this assessment.

Note: Extension is not available for this assessment for the reason of in-person viva. Students are advised to apply EC (if they have a EC case) if they could not meet the due date.

RESOURCES & CONTACT

Resources

- Approved dataset: Zenodo record 11066414
- Transformers.js documentation
- Transformers.js pipeline tasks
- Suggested browser model: Xenova/all-MiniLM-L6-v2
- Ollama model library
- Suggested local model: Gemma 3
- Ollama API
- Ollama structured outputs

Note: *Model availability and exact file sizes can change. Record the identifiers and versions actually used, and ensure the assessor can reproduce your setup from the submitted README.*

Contact Information:

Assessment Setter: Cheng.C.Wang@uea.ac.uk

EVALUATION & GRADING

The marking scheme is outlined below:

Essential Features (50%)

Criterion	Feedback	Mark
Application and data journey (10 marks)		
Transformers.js semantic feature (20 marks)		
Bounded local-agent design (15 marks)		
Reliability, usability and code quality (5 marks)		

Desirable Features (25%)

Feedback	Mark
	/25

Oral Defence (25%)

Feedback	Mark
	/25

PLEASE READ ALL THE INFORMATION BELOW CAREFULLY

ACADEMIC INTEGRITY

The University takes academic integrity very seriously. You must not commit plagiarism, collusion, or contract cheating in your submitted work. Our Policy on Plagiarism, Collusion, and Contract Cheating explains:

- what is meant by the terms 'plagiarism', 'collusion', and 'contract cheating'
- how to avoid plagiarism, collusion, and contract cheating
- using a proofreader
- what will happen if we suspect that you have breached the policy.

It is essential that you read this policy, and you undertake (or refresh your memory of) our school's training on this. You can find the policy and related guidance here:

https://my.uea.ac.uk/departments/learningand-teaching/students/academic-cycle/regulations-and-discipline/plagiarism-awareness

In this assessment, working with others is **NOT PERMITTED**. All aspects of your submission, including but not limited to: research, design, development and writing, must be your own work according to your own understanding of topics. Please pay careful attention to the definitions of contract cheating, plagiarism and collusion in the policy and ask your assessment setter if you are unsure about anything.

AI POLICY AND USE

To ensure fairness and clarity, this module uses a Traffic Light system to outline exactly how you can and cannot use generative AI tools for your assessment.

🔴 RED	🟡 AMBER	🟢 GREEN ✓

🟢 GREEN: FULL AI INTEGRATION ENCOURAGED

Generative AI tools are permitted and/or are a core component of this assessment.

Permitted Uses: You may use AI tools to assist with debugging.

Declaration Requirement: *You must include an AI_USAGE.md file detailing which tools were used and how outputs were integrated. You remain fully responsible for the accuracy of any AI-generated content.*

APPROVAL

Signed off at version 3 of this brief.

Checker: Check completed by Dr Jeannette Chin on 2 November 2026

Cluster lead: Optional sign-off, not completed